Summer of Code 2026: Machine Learning Applications in Nature and Finance

Aaditya Vichare

2026

1 Introduction

Machine learning is useful wherever we see repeated patterns, noisy measurements, or hidden structure in data. In this report, supervised learning is used for prediction with labeled examples, while unsupervised learning is used for discovering patterns without labels.

2 Problems in Nature

The table below lists five common problems seen in nature and suitable learning methods for each.

Nature problem	Task type	Best model idea	Benefits
Animal species identification from images	Supervised classification	CNN, ResNet	High accuracy, robust feature learning, useful for biodiversity studies and conservation.
Leaf disease detection from images	Supervised classification	CNN, transfer learning	Early diagnosis, faster crop protection, lower manual inspection effort.
Rainfall or temperature forecasting	Supervised regression	Linear regression, Random Forest, XGBoost, LSTM	Supports planning, captures trends, helps in climate and agriculture decisions.
Soil or water quality grading	Supervised classification/regression	Random Forest, SVM	Handles mixed features, gives practical environmental monitoring.
Wildlife camera-trap grouping	Unsupervised clustering	K-means, DBSCAN, hierarchical clustering	Finds hidden groups without labels, useful when manual tagging is expensive.

For supervised tasks, CNNs work best for image-based nature problems because they learn spatial patterns automatically. For unsupervised tasks, DBSCAN is often strong when natural data contains noise and irregular clusters.

3 Stock Regression

Regression is suitable when the target is continuous, such as stock close price, adjusted close price, daily return, or volatility. A simple example is predicting next-day closing value using historical prices and derived indicators.

Target variable	Example inputs	Comment
Close price	Open, high, low, volume, moving averages	Most direct regression target.
Adjusted close	Same as above plus splits/dividends context	Better for long-term analysis.
Next-day return	Past returns, momentum, RSI, MACD	Often more stable than raw price.
Volatility	High-low range, intraday return statistics	Useful for risk modeling.

Single linear regression is usually not fine for stock price prediction because stock movement is nonlinear, noisy, and affected by many variables. A better feature space for multiple linear regression includes lagged prices, moving averages, returns, volume, volatility, technical indicators such as RSI and MACD, market index returns, and possibly sector or macroeconomic variables.

4 Gradient Descent Beyond Neural Networks

Gradient descent is an optimization method that reduces an objective function step by step. It is widely used outside neural networks in linear regression, logistic regression, support vector machines, matrix factorization, and portfolio optimization.

In linear regression, gradient descent minimizes mean squared error and updates parameters by moving opposite to the gradient. In logistics or classification, it minimizes cross-entropy or another loss function. In finance, it can help optimize a model fit or tune portfolio weights by minimizing risk-adjusted cost.

Backpropagation is the efficient application of the chain rule to compute gradients in layered computations. The same idea can be used in any differentiable model with a computation graph, not only neural networks. For example, in a physics-informed model or differentiable simulation, gradients can be propagated backward through intermediate formulas to update parameters.

5 Why These Methods Work

Supervised learning is best when correct answers already exist, such as labeled species images or historical stock values. Unsupervised learning is best when the goal is to discover structure, such as hidden animal groups or environmental clusters. Gradient-based optimization is useful whenever the model is differentiable and parameters must be tuned efficiently.

6 Conclusion

Nature and finance both generate complex data with noise, trends, and hidden relationships. Supervised learning helps prediction, unsupervised learning helps discovery, and gradient descent provides the optimization engine that makes many models trainable.